William Lorenzo

Machine Learning Engineer | Applied AI | ML Systems

San Francisco Bay Area, CA | (650) 255-1468 | ssfjdm@gmail.com

Portfolio | GitHub | LinkedIn | ResearchGate

Profile

Applied ML engineer and researcher building systems across **GPU-accelerated machine learning, computer vision, clinical informatics, and cloud data infrastructure**. Experience developing deep learning models, AWS data pipelines, and reproducible ML systems with PyTorch, TensorFlow, SQL, and NVIDIA CUDA, supported by rigorous training in statistics and experimental analysis.

Technical Skills

ML & AI: PyTorch, TensorFlow, Scikit-Learn, LSTM, Convolutional Neural Networks, Feedforward Neural Networks, NLP, NVIDIA CUDA

Programming & Data: Python, SQL, R, C++, Pandas, NumPy, SciPy, MySQL

Cloud & Systems: AWS (S3, CloudWatch, EC2), Lambda, Terraform, Azure, ETL, Data Pipelines, Linux, Git

Statistics & Analytics: Regression, Classification, ANOVA, Hypothesis Testing, Statistical Inference, Feature Engineering, Model Evaluation, Tableau

Experience

Interlinked — AI-Powered Wildfire Intelligence Platform **Berkeley, CA**
AI Engineer & Data Analyst *May 2025 – May 2026*

- Developed physics-informed wildfire forecasting workflows combining **Rothermel and level-set methods** with machine learning surrogates for disaster-response intelligence.
- Implemented **U-Net, ConvLSTM, and Graph Neural Networks** for spatiotemporal forecasting using MODIS/VIIRS observations, wildfire perimeters, vegetation, terrain, and historical incidents.
- Built model-ready geospatial datasets and automated **AWS S3, Lambda, and CloudWatch** pipelines for ingestion, validation, monitoring, and quality assurance.

Education

University of California, San Diego **La Jolla, CA**
Master of Data Science (MDS) *Jan 2027 – Jun 2028*

California State University, East Bay **Hayward, CA**
B.S. Statistics — Data Science Concentration *Aug 2021 – May 2026*

Magna Cum Laude | GPA: **3.81** | Dean's List: 2024–2026

Relevant Coursework: Statistical Inference, Probability, Regression, ANOVA, Statistical Learning, Python for Data Analysis, R for Data Science, Data Wrangling & Pipelines, Database & SQL, Data Visualization

Honors: CSUEB Honors Convocation – Magna Cum Laude (2026) · SCLA (2025) · Tau Sigma (2025) · NSLS (2024)

Selected Research & Engineering

GPU-Accelerated ML Benchmarking with NVIDIA CUDA and Nsight **June 2026**
Python | CUDA | PyTorch | TensorFlow | XGBoost | NVIDIA Nsight Systems

- Built a reproducible benchmark on a **581K-row dataset** comparing CPU/GPU training and inference across Random Forest, XGBoost, TensorFlow, and PyTorch.
- Profiled runtime, GPU utilization, memory, and bottlenecks with Nsight Systems; measured workload-dependent acceleration up to **26.92**×.

Computational Analysis of UCSF Clinical Trial Data Using MySQL, NLP, and Machine Learning **May 2026**
Python | MySQL | SQL | NLP | TF-IDF | Machine Learning

- Designed an end-to-end MySQL/Python workflow for UCSF-affiliated ClinicalTrials.gov data; applied SQL analytics, NLP, classification, and clustering to characterize trial patterns and predictive signals.

Myocardial Perfusion Segmentation with PyTorch U-Net **March 2026**
PyTorch | U-Net | ResNet34 | DICOM | NIfTI | Computer Vision

- Built a ResNet34-based U-Net medical imaging pipeline achieving approximately **0.86 validation Dice**; evaluated augmentation, optimization, class imbalance, and model generalization.